

Democratic Resilience in the Twenty-First Century: Search for an Analytical Framework and Explorative Analysis

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Abstract

Why are some democracies more resilient than others? How can their resilience be improved? Answering these questions requires valid conceptualization and reliable measures of democratic resilience. This study presents a novel conceptualization and measurements of democratic resilience. It differentiates between democratic resilience as regime performance and regime capacity and introduces the Democratic Resilience Capacity Index. The index is the first and only measure available to empirically compare the ability of democracies to become or to remain resilient. We demonstrate the usefulness of our approach for empirical research through an exploratory study of 103 democracies since 2000. We find that resilience capacity strengthens the ability of a democracy to prevent a substantial loss of its democratic qualities and regime breakdown, but not its ability to recover from autocratization. This suggests that different forms of democratic resilience, though temporally connected, depend on different forms or combinations of resilience capacities.

Keywords

democratic resilience, resilience measurement, resilience capacity, resilience performance, democratic backsliding, autocratization, Democratic Resilience Capacity (DRC) Index

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Introduction

Worldwide, concerns about the health of democracy shape academic and public discourses. While global democracy reports stress the pressing need to act for democracy to prevail (Economist Intelligence Unit, 2025; Freedom House, 2025; Nord et al., 2025a),

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scholars have called for a better understanding of what protects democracies and enables political systems to cope with threats to their democratic institutions, norms and practices (Merkel and Lührmann, 2021; Riedl et al., 2024). Although democracy research has developed useful concepts and tools to analyze modes and outcomes of democratic regression, the search for explanations of these modes and outcomes is still at an early stage. In this context, the concept of democratic resilience has increasingly gained currency, although it has rarely been conceptualized or measured coherently (cf. Holloway and Manwaring, 2023; Walz et al., 2025). Given the concerns about the fate of democracy in both industrialized and in developing societies, thinking about democratic resilience in theoretical, conceptual and comparative terms is one of the most pressing challenges in (de-)democratization studies.

Despite the new interest in the concept of resilience, political science generally lacks thorough conceptualizations and, especially, empirical indicators to identify and to measure different forms and levels of democratic resilience. Moreover, in order to better understand why some democracies are more resilient than others, under which conditions they resist or recover from stress and shocks, and how the resilience of democratic procedures, processes and principles can be strengthened, it is necessary to differentiate between levels or degrees of democratic resilience on one hand, and the features which enable democratic regimes to achieve or remain a certain level of resilience. Answering these questions requires valid conceptualization and reliable measures of democratic resilience. This study presents a conceptualization and measurements of democratic resilience. First, we present a novel conceptualization of democratic resilience that differentiates between resilience performance and resilience capacities of political regimes. Second, we combine different approaches to measuring autocratization to capture three varieties of democratic resilience performance: maintenance, resistance and bounce-back resilience. Third, building on the novel differentiation between democratic resilience as regime performance and regime capacity, the study introduces the Democratic Resilience Capacity (DRC) Index. The DRC Index is the first and only measure available to empirically compare the ability of democracies to become or to remain resilient. Fourth, we demonstrate the usefulness of our framework in a two-step analysis. First, we measure resilience capacities in up to 117 countries. Next, we examine the possible relationship between resilience capacities and the three varieties of resilience performance in 103 democratic regimes since 2000. Our statistical analysis demonstrates that resilience capacity as measured by the DRC Index is positively associated with the ability of a political regime to prevent a substantial deterioration of the democratic qualities of its political institutions and processes and, if it experiences democratic erosion, to resist the breakdown of its democratic institutions. However, we cannot confirm that democratic resilience has a substantial effect on the ability of a political regime to bounce back from an episode of autocratization. This suggests that it is necessary to differentiate between resilience capacity and resilience performance and between different but temporally connected forms of democratic resilience. Furthermore, different types of democratic resilience depend on different forms or combinations of resilience capacities. The novel conceptualization, original indicators and fresh insights into the relationship between different types of democratic resilience and various resilience capacities offers a new starting point for future efforts to identify pathways toward resilient democracy as well as for evidence-based efforts to strengthen the resilience of democracies at risk.

Defining Democratic Resilience

For many years, resilience has been an established concept in many academic disciplines (Chandler and Coaffee, 2016; Rogers, 2015). However, in political science, considering democratic systems through the lens of resilience is a relatively new perspective, which has gained attention mainly in research on current threats to democracy—autocratization and democratic backsliding in particular (Boese et al., 2021; Croissant and Waldner, 2025; Holloway and Manwaring, 2023; Merkel and Lührmann, 2021; Riedl et al., 2024; Volacu and Aligica, 2023; Walz et al., 2025). Nonetheless, democratic resilience remains a term with no standardized definition or operationalization, which carries a range of different meanings and entails a variety of different practices in different national contexts.

We generally understand resilience as the ability of an entity or system to resist shocks, to absorb them, to bounce back from them and to move forward, in order to maintain or enhance its identity, if not its structures and functions (Folke et al., 2004: 558; Stollenwerk et al., 2021). As Holloway and Manwaring (2023: 72) note, “resilience is . . . not a single property that a democratic system would possess and exhibit in all circumstances, but rather a response process of ‘patterned adjustments’ that a system (or agents within) may demonstrate in *varying degree*” (emphasis added).

More specifically, Boese et al. define *democratic* resilience as “the ability to prevent substantial regression in the quality of democratic institutions and practices” (2021: 886). This parsimonious definition has the disadvantage that it refers exclusively to the performance resilience of democratic regimes. Sorsa and Kivikoski (2023: 2) understand democratic resilience as “the capacity of democratic institutions and practices to absorb and recover, adapt, innovate, or transform in response to shock or crisis.” The problem with the scope of this definition of democratic resilience is its emphasis on “the capacity of democratic institutions and practices.” However, one of the critical distinctions we draw in this study is between democratic resilience as performance and as ability. If the general definition is phrased in terms of ability, however, it cannot capture the idea of democratic resilience as performance. Merkel and Lührmann (2021: 874) propose a broad concept of democratic resilience as

the ability of a democratic system, its institutions, political actors, and citizens to prevent or react to external and internal challenges, stresses, and assaults through one or more of the three potential reactions: to withstand without changes, to adapt through internal changes, and to recover without losing the democratic character of its regime and its constitutive core institutions, organizations, and processes

Although Lührmann and Merkel do not conceptualize democratic resilience in terms of regime performance and regime capabilities, their definition encompasses both accounts of resilience—as performance and as capacity of democratic systems. Inspired by Bănică et al. (2021) as well as Volacu and Aligica (2023), we make an analytical distinction between how resilient a democratic system actually *is* (resilience as “performance”) and the ability of that system to *remain or to become* resilient in terms of sustaining or enhancing its democractiness (resilience as “capacity”).¹ We define *resilience capacities* as those abilities of a political system that help democratic procedures, principles and practices to maintain their given achievement levels, to resist erosion or breakdown and to develop sufficient elasticity to bounce back from regress. Resilience capacity implies identifying and measuring the conditions or driving factors that are hypothesized to position the political system to respond and recover better. The resilience

capacity operates as a moderator that determines the extent to which a given stressor poses a threat to that political system (Walz et al., 2025).

Resilience as performance manifests itself through one or more of the following three outcomes:

1. *Maintenance*: a democracy withstands the given level of stress without substantial and significant regression of its democratic quality or regime architecture. This means that the democratic quality of practices, principles and processes continue on the same or similar level during a period of stress or crisis as before and after that period of time.
2. *Resistance*: a democratic system experiences an episode of democratic erosion but is able to limit the negative impact of a shock or crisis so that it can persist in diminished form and persevere what remains of its democratic architecture and avoid the breakdown of its core identity as a (minimally) democratic regime.
3. *Bounce-back*: This kind of resilience performance is manifested in the use of stress-absorbing strategies and mechanisms that allow a political system, once democratic regression has come to an end at a lower level of democratic quality than before the onset of the episode, to recover from losses of democratic qualities and to achieve democratic improvement, perhaps beyond the pre-crisis level.

The difference between democratic resilience as regime maintenance and resistance mirrors the two-stage concept of democratic resilience introduced by Vanessa Boese and her co-authors (2021). They contend that a democracy can be resilient by avoiding the onset of autocratization, and, failing that, a democracy can also be resilient by avoiding the breakdown of democracy in the second stage. This approach is an important first step toward developing valid and reliable measures of democratic resilience, though it does not accommodate the full notion of resilience as “bouncing back” from initial damage.

Against this backdrop, Marina Nord and her co-authors recently introduced their concept of “democratic turnarounds” by which they mean processes of democratization that follow an initial erosion of democracy and are, in one way or the other, endogenously interlinked and closely connected in time to prior incidents and processes of autocratization (Nord et al., 2025b). Such episodes of democratic turnaround are cases of reversed regression or “bounce back” which can result in a restoration of the regime’s previous level of democratic qualities, a substantial improvement in democratic traits compared to the previous level, or an increase in democratic qualities, though the new democratic level constitutes a substantial decline compared to the previous level.

Our conceptualization of democratic resilience combines the approaches introduced by Boese et al. as well as Nord and their co-authors, but goes beyond these proceduralist understandings of democratic resilience as performance by introducing the concept of resilience capacity.

The capacity of a system to be resilient and the level of actual resilience are linked by the concept of resilience mechanisms—those causal processes through which capacities impact the actual resilience performance of the system (Figure 1).

These mechanisms may play out in different ways. Democratic resilience includes the state as one governance actor among many, but also non-state and societal actors in a broader sense. If democracies are resilient, they are not only able to fend off stressors and crises, but also to develop coping mechanisms allowing them to deal with risks in the long run, preserving, reviving or enhancing their level of “democraticness.” A comprehensive

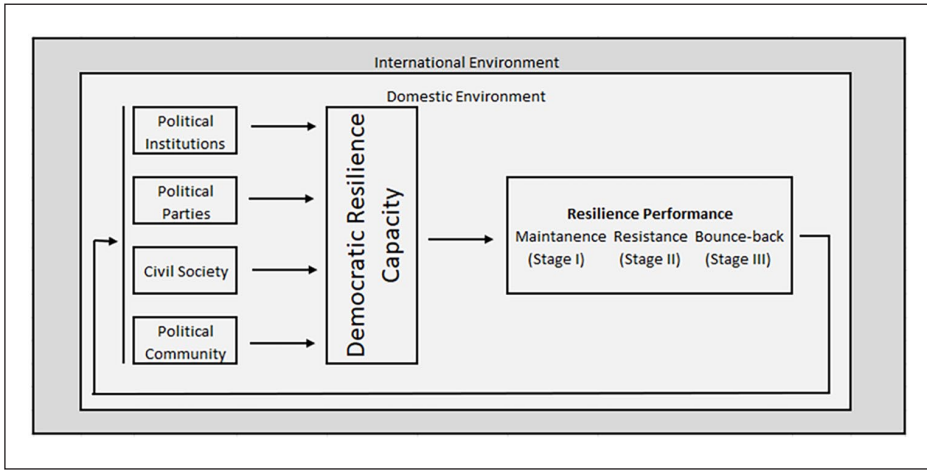


Figure 1. Theoretical Framework of Democratic Resilience.

conceptualization of the conditions, capacities and consequences of democratic resilience would include resilience mechanisms (see also Lührmann et al., 2020). However, given the exploratory character of this article, and our main research interest detailed in the introduction, the rest of the paper focuses on resilience performance and resilience capacities.

Measuring Democratic Resilience

Accurately measuring democratic resilience requires reliable and valid data on democracy. Numerous indicators of democracy can be utilized to measure democracy across a large number of cases (Møller and Skaaning, 2023). In contrast, there is a lack of well-tested metrics for measuring resilience. There is only one attempt to systematically measure the democratic resilience of a large number of political systems across different regions and time periods (Boese et al., 2021). Performance indicators of democratic resilience must be able to measure the level or degree of resilience a system exhibits at a given point in time, whereas capacity indicators identify a set of measurable characteristics within a given country over a period of time that influence the level of democratic resilience.

Measuring Democracy

Democracy barometers differ in terms of their understanding of democracy and the type of data to be used. Despite the nature of democracy as an “essentially contested concept” (Collier et al., 2006), actual empirical research relies on procedural understandings of democracy, that is, democracy is defined in terms of political rules and procedures, not in terms of its outcomes (i.e. “social democracy” or “strong democracy”; see Bogaards, 2010; Coppedge et al., 2020; Held, 2006). The choice of a specific concept of democracy, in turn, has important consequences both in terms of how one measures the democratic qualities of political regimes in general and with regard to the resilience of a democracy in particular. Existing democracy barometers and studies on democratic regression and

resilience generally adopt electoral or liberal conceptions of democracy (cf. Boese, 2019; Boese et al., 2021; Riedl et al., 2024). Electoral democracy is an example of a concept of democracy in which only the minimum procedural requirements of a democracy are met, which are necessary for civic participation in the political process through voting and opinion making. A liberal democracy, in turn, fulfills all the conditions of electoral democracy, but goes beyond them by including procedures of rule of law, minority protection and horizontal accountability, among others (Coppedge et al., 2020).

Another long-standing debate among democracy researchers concerns the question of whether measures of democracy should rely on expert-coded indicators, observational data, individual-level survey data, or a combination of different types of data (Fuchs and Roller, 2018; Geissel, 2023; Skaaning, 2018). In the context of current research on democratic backsliding, this debate has once again attracted a great deal of attention (*Political Science & Politics* 2024). Most democracy barometers exclude citizen perceptions as identified in opinion poll data because these types of data can be problematic if scholars plan to compare democratic qualities across countries or globally. Instead, most democracy barometers rely on expert-coded data (Boese, 2019). Such expert-coded data may be potentially vulnerable to issues of inter-coder reliability and coder bias, and information that expert coders rely on may be biased to start with (Little and Meng, 2024: 159). However, relying solely or mainly on “more objective data” (Little and Meng, 2024) is insufficient to capture the complexities that characterize the concepts of democratic quality and democratic resilience. In the sense of the evaluation scheme for democracy indices proposed by Munck and Verkuilen (2002), the “reliability” criterion may be overemphasized, while the “validity” aspect is neglected. Moreover, it is an illusion that more “objective” indicators are always observer-invariant (Knutsen et al., 2024; Lott and Croissant, 2024).

Democracy barometers exist in large numbers, but the Varieties of Democracy (V-Dem) dataset is the most ambitious and methodologically advanced one. The V-Dem project distinguishes between five high-level principles of democracy (electoral, liberal, participatory, deliberative and egalitarian), operationalizes these principles and collects data to measure those (Coppedge et al., 2020). Of these principles of democracy, the electoral democracy and the liberal democracy indices are most widely used in empirical research. With about 400 specific indicators, multiple, independent coders for each (non-factual) question, systematic inter-coder reliability tests, confidence bounds for all point estimates associated with non-factual questions and transparent aggregation procedures, the V-Dem Project is the de facto gold standard in empirical democracy research.

Measuring Resilience Performance

Measuring democratic resilience is a challenging endeavor. Generally, one can distinguish between two approaches. One approach measures democratic resilience performance as the extent of change in the level of democratic quality above a defined threshold over a defined period of years that is long enough to exclude ephemeral changes (Lott and Croissant, 2024). While a substantial drop in the level of democratic qualities of a political regime would indicate a weaker resilience performance of a system, only ephemeral changes or a substantial increase would suggest a stronger resilience performance by the system.

However, this approach comes with some pitfalls and tradeoffs. The first is to define a time span that is long enough to exclude ephemeral changes in the level of democracy and short enough not to miss wave motion (in terms of short-term democratic regression and

fast recovery). The second is setting a threshold to detect substantial declines in the level of democracy that is neither undemanding nor over-demanding (Coppedge, 2017: 7). Using a larger time interval and an undemanding threshold, researchers are likely to overestimate autocratization (Type 1 error—false positive). Type II errors (false negatives) are likely in cases where researchers choose a short time lag (for example 1 year) and apply a demanding threshold to detect substantial declines in the level of democracy. Overall, setting the time lag and thresholds remains an error-prone decision, which is why we decided against following this approach.

An alternative approach—the “episode approach”—treats democratic resilience as the avoidance of a (gradual) process of connected years during which sustained declines in the quality of democracy take place. In recent years, this approach has become increasingly popular in empirical democracy research. Maerz et al. (2024a) have made a key contribution to the measurement of autocratization with the Episodes of Regime Transformation (ERT) dataset.² Autocratization episodes are conceptualized as interrelated periods of substantial cumulative decline in democratic qualities in any given political regime. The episode approach treats democratic resilience as a process of connected years during which a political regime is able to avoid a substantial loss in democratic quality or a regime transition from a more democratic to a less or non-democratic regime (cf. Boese et al., 2021).

This approach is—in our view—an important first step toward developing valid and reliable measures of democratic resilience, though it emphasizes notions of democratic resilience as “resistance” and the continuation of the discrete regime type, but it does not accommodate the full notion of resilience as a “bouncing back” from initial damage, or the improvement of democratic qualities above pre-existing levels. The concept of “democratic turnarounds” measures processes of democratization that follow initial erosion of democracy and are in one way or the other endogenously interlinked and closely connected in time to prior episodes of autocratization (Nord et al., 2025b). Such episodes of democratic turnaround are cases of “bounce-back” which can result in a restoration of the regime’s previous level of democratic qualities (“U-shaped” turnaround), a substantial improvement in democratic traits compared to the previous level (“J-shaped”) or a (substantial) increase in democratic qualities, though the new democratic level constitutes a substantial decline compared to the previous level (“L-shaped”) (Nord et al., 2025b). We combine the two conceptual approaches by Boese et al. (2021) and Nord et al. (2025b) into a three-stage approach to democratic resilience as mentioned, using their operationalization of autocratization onset in an existing democracy, democratic breakdown and episodes of democratic turnaround as approximations of our three forms of resilience performance in democracies.

Measuring Resilience Capacity

Resilience capacity assesses the capacity of a polity to resist and bounce back from a crisis. If the resilience capacity matches the intensity of a stressor, then a political system could cope with the stress without significant changes in its quality, would be able to escape slow or sudden death, or bounce-back from an initial decline in the quality of democracy. When the level of stress far exceeds the resilience capacities, then a decline or breakdown of democracy seems to be the most likely outcome. In this case, a system’s short-term ability to build up new resilience capacities toward the stressor is of critical importance for achieving a bounce-back scenario.

The scientific literature uses an enormous number of indicators as determinants of democratic breakdown and survival.³ Socioeconomic factors, while often linked to democratic stability, show mixed effects: affluent democracies tend to be more stable, but the relationship between economic performance, income and democratic backsliding is context-dependent (Knutsen and Teorell, 2024; Treisman, 2020). Global economic crises like the great recession of 2007 to 2009; natural or human-made disasters such as the COVID-19 pandemic; and foreign threats like the Russian invasion of Ukraine are exogenous shocks that can trigger endogenous reactions that decrease or increase the level of democratic qualities of a political regime, but do not constitute a resilience capacity as such. Similarly, the strength of state capacity (Andersson and Teorell, 2024), the intensity of distributive conflicts (Haggard and Kaufman, 2016; Houle, 2024), ethno-religious fractionalization and conflicts (Maerz et al., 2024b), or authoritarian legacies (Morlino, 2024) are contextual conditions that, through the behavior and responses of relevant political elites, affect the chance that a democracy is forced to demonstrate a resilience response, but these factors do not constitute a form of resilience capacity.

Our conceptualization of resilience capacity builds on a four-level model recently proposed by Merkel and Lührmann (2021), which is also informed by democratization literature. The four levels are the following:

1. macro-institutional (core institutions of the democratic regime);
2. political parties;
3. societal (civic culture and civil society); and
4. political community.

However, Merkel and Lührmann's framework lacks a concrete operationalization of these four levels. We fill this gap by identifying ten indicators of resilience capacity. Table 1 summarizes the variables, indicators and data sources and depicts the direction of the relationship between capacity and performance.

ad 1) The macro-institutional level concerns core procedural rules and institutions which are relevant to the survival and democratic quality of the regime. Acknowledging that incumbent-led backsliding is the dominant mode of democratic regression in the post-Cold War era (Kendall-Taylor et al., 2019: 275), scholars agree that executive constraints can make a democratic system more resilient. Although autocratizing governments have increasingly turned to the judiciary to do their dirty work (Landau and Dixon, 2020), judicial constraints on the executive are commonly seen as *potential* bulwarks against incumbent-led backsliding and as institutional guardrails in times of democratic regression (Boese et al., 2021). However, executive or horizontal constraints are conceptually broader than judicial review. The institutional relationship between the executive on the one hand and the legislative branch of government and between government and non-judicial watchdog agencies also affects the quality of the "horizontal accountability mechanism" (Lührmann et al., 2020). We assume that democracies have a stronger capacity to be resilient when the executive is effectively constrained by an independent and impartial judicial and legislative oversight.

Moreover, Gerring et al. (2005) demonstrate that the impact of the accumulated experience with democratic institutions and practices within a polity increases over time (see also Edgell et al., 2020; Svolik, 2008). "Democracy stock," therefore, is another critical component of resilience capacities: political regimes with more and stronger democratic

Table 1. Variables and Indicators of Democratic Resilience Capacity.

Variables	Justification for inclusion	Indicators and data sources
Macro-institutional level		
Democracy Stock	Political systems with more and stronger democratic experiences and legacies have a higher capacity to anticipate, adopt, resist or recover from the repercussions of external shocks or stress.	EDI (Coppedge et al., 2024); cumulative weighted sum of EDI values for all previous years. A conventional two and a half-percent annual depreciation rate (1 - δ) was used.
Horizontal Accountability (V-Dem)	The more the executive is constrained by independent judiciary and effective legislative oversight, the stronger is horizontal accountability, which is an important resilient mechanism. In addition, the stronger the compliance with judiciary and the higher the court independence, the stronger the incentives for political and economic actors to keep and defend democracy.	Eight indicators from the Horizontal Accountability Index (High court independence, Lower court independence, Compliance with high court, Compliance with judiciary, Executive respects constitution, Executive oversight, Legislature investigates in practice, Legislature questions officials in practice). Higher scores indicate stronger constraints (Coppedge et al., 2024, 296ff.).
Political actors		
Anti-pluralist Party Index (V-Party)	The more political parties are committed to pluralism and democratic processes, the better the ability of the party system to reduce political uncertainty and to provide more stable pro-democratic representation.	Democratic Party Index (Angiolillo et al., 2025; V-Dem; Coppedge et al., 2024); higher scores indicate more anti-pluralist party preferences in the party systems and thus lower resilience capacity.
Polarization (V-Dem)	The stronger political polarization, the more a society is divided into antagonistic political camps and the weaker is the resilience capacity of the political system.	Political polarization indicator (V-Dem; Coppedge et al., 2024); higher values indicate more polarization.
Political Violence (V-Dem)	The more political violence, the weaker is the resilience capacity of the political system.	Political violence indicator (V-Dem; Coppedge et al., 2024); higher values indicate more political violence.
Civic culture and civil society		
Robustness of civil society	The more robust a civil society, the higher the capacity for vertical accountability, public consultation and consensus-building and critical support of the state by society.	Indicators from the core civil society index (v2xcs_ccs) by V-Dem (CSO entry and exit; CSO repression; CSO participatory environment); higher scores indicate better resilience capacity (Coppedge et al., 2024).
Distribution of power resources	Wider distribution of relevant power resources make it easier for citizens to play a role in democratic politics which strengthens democratic resilience.	Indicators from the equal Access Index (power distributed by gender; power distributed by socioeconomic position; power distributed by social group), which measure the degree to which all groups “enjoy equal de facto capabilities to participate, to serve in positions of political power, to put issues on the agenda, and to influence policymaking.” (Coppedge et al., 2024: 59).
Political community		
Political trust in representative institutions	Higher levels of political trust in representative institutions induces less openness for anti-system alternatives and promote willingness of actors to overcome collective action problems and cooperate in the face of emerging or present risks to democratic systems.	Valgarðsson et al. (2025) latent estimates of trust in the government and trust in the parliament (Different surveys, such as ESS, WVS, EVS, LB, ISSP, etc.) based on latent variable approach by Claassen (2019). Higher values indicate more political trust and thus more resilience capacity.
Political trust in order institutions	Higher levels of political trust in order institutions induces less openness for anti-system alternatives and promote willingness of actors to overcome collective action problems and cooperate in the face of emerging or present risks to democratic systems.	Valgarðsson et al. (2025) latent estimates of trust in the police and legal order (Different surveys, such as ESS, WVS, EVS, LB, ISSP, etc.) based on latent variable approach by Claassen (2019). Higher values indicate more political trust and thus more resilience capacity.
Satisfaction with Democracy	Higher confidence in democracy induces less openness for anti-government/anti-system alternatives.	Claassen (2020) latent estimates of confidence in democracy (Different surveys, such as ESS, WVS, EVS, LB, ISSP, etc.).

experiences and legacies have a better capacity to anticipate, adapt, resist or recover from exogenous shocks and endogenous stress.

ad 2) At the level of relevant political actors, political parties play a key role (Bernhard et al., 2020; Merkel and Lührmann, 2021). To operationalize this level, we identify three party-related variables. First, democracy relies on nonviolent mechanisms of political participation and political inclusion (Rummel, 2002). Political party systems with stronger commitment to democracy and pluralism strengthen the resilience capacity of a democracy by protecting its against anti-democratic political forces. In contrast, political systems in which relevant political parties promote anti-pluralist ideology, and whose supporters generally interact in a hostile manner, have a weaker resilience capacity (Coppedge et al., 2022; Medzihorsky and Lindberg, 2024; Riedl et al., 2024).⁴ Second, political polarization threatens the ability of citizens to find a common procedural consensus which democracy rests upon and motivates citizens to overlook anti-democratic behavior by those who represent their policy preferences (McCoy et al., 2018; Sartori, 1976; Svobik, 2018). Therefore, we assume that the stronger political, and especially affective polarization in a political system, the weaker its resilience capacity against democratic regression. Third, political violence is incompatible with democratic processes and opinion making. However, some democracies regularly face political violence from non-state actors that undermine the democratic process. Therefore, we assume that the greater political violence in a political system, the weaker its resilience capacity against any forms of democratic regression.

ad 3) We approximate this capacity level via two indicators. First, the robustness of civil society. While much of the literature on civil society and democratic regressions focuses on the role of civil society as a resistance actor, we treat civil society as a resilience capacity: The more widespread and anchored democratic values and attitudes are in a society, and the more robust a civil society, the better protected democracy is to external shocks and internal challenges (Bernhard et al., 2020; Merkel and Lührmann, 2021). Second, the resilience capacity of a democracy ought to be stronger when social groups “enjoy equal *de facto* capabilities to participate, to serve in positions of political power, to put issues on the agenda, and to influence policymaking” (Coppedge et al., 2024: 59; see also Vanhanen, 2000).

ad 4) The political community of citizens concerns a fourth component of resilience capacity. As Merkel and Lührmann (2021) note, early works defined democratic resilience in terms of the members of a political community’s attachment to democratic ideals. The greater the proportion of citizens that shares this identity the easier it is for democracy to function and persist. Furthermore, citizens’ common sense of belonging and a shared citizenship agreement strengthen civic allegiance to democracy and render it more difficult for anti-system alternatives to gain political traction (Riedl et al., 2024; Somer et al., 2021). We approximate this fourth dimension of the resilience capacity of a democracy through five survey-based indicators: trust in representative institutions (two indicators), trust in order institutions (two indicators) and confidence in democracy (one indicator). Trust is one of the most important factors that contributes to the citizens of a society attaining a sense of community, and political trust is related to the confidence that political institutions will act in their interest. Inglehart (1988) notes that societies with high levels of interpersonal trust and satisfaction with life and politics are more likely to be stable democracies. We assume that higher trust in “representative” and “implementing” institutions⁵ among the citizens strengthens the ability of regimes to respond to shocks and stress and renders it more difficult for anti-system alternatives to gain political

Table 2. Conceptual Alignment Across Components (Bayesian Confirmatory Factor Analysis).

Dimension	Indicator	Loading (Λ)	Uniqueness (Ψ)
Macro-institutional	Democratic stock	0.76	0.418
	High court independence	0.88	0.233
	Lower court independence	0.87	0.244
	Compliance with high court	0.87	0.246
	Compliance with judiciary	0.88	0.221
	Executive respects constitution	0.82	0.327
	Executive oversight	0.82	0.334
	Legislature investigates in practice	0.82	0.332
	Legislature questions officials in practice	0.70	0.527
Political actors	Polarization	0.71	0.496
	Political Violence	0.75	0.435
	Anti-Pluralist-Party Index	0.46	0.795
Civil society and civic culture	Power distributed by gender	0.64	0.586
	Power distributed by socioeconomic position	0.74	0.457
	Power distributed by social group	0.56	0.691
	CSO entry and exit	0.88	0.232
	CSO participatory environment	0.89	0.206
	CSO repression	0.74	0.445
Political community	Trust in representative institutions: parliament	0.85	0.249
	Trust in representative institutions: government	0.79	0.349
	Trust in order institutions: police	0.65	0.572
	Trust in order institutions: legal system	0.85	0.279
	Satisfaction with Democracy	0.63	0.605

For the Democratic Resilience Capacity Index construction, the indicators for political parties dimensions were inversely rescaled to obtain an index that ranks from low capacity (0) to high capacity (1).

Source: see Table 1.

traction. In addition, the more citizens and elites have satisfaction in their democracy the more they are willing to comply with democratic decisions and policies. This reduces reaction and implementation costs for the democratic system and strengthens its capacity to deal effectively with external or internal disturbances. We use Bayesian factor analysis (BFA) to assess the content validity of the different components of our democratic resilience capacity measure.⁶ Table 2 tests whether the different indicators reflect one respective underlying systematized concept (e.g. macro-institutional dimension). We implemented a BFA to incorporate measurement error in the latent variables we used to construct the different dimensions, into the model. All of our indicators are manifest variables, which were estimated using Bayesian methods. Compared to the traditional frequentist approach, BFA assumes the same likelihood function, but uses prior information to “overcome the rotational invariance problem, typically by assuming, a priori, that at least one loading is strictly positive (or negative)” (McMann et al., 2022, 433). However, the factor loadings from the BFA can be interpreted in the same way as factor loadings from frequentist methods.

We start with 1800 simulated draws from the posterior distribution of each latent variable.⁷ In a second step, we fit the Bayesian factor model to each draw, using Markov chain Monte Carlo (MCMC) methods and vague priors to simulate 900 “draws from the

posterior of the factor model for that set of manifest variable draws” (McMann et al., 2022, 433).⁸ In the last step, we combine the posterior draws from every run of the factor model, and compute parameter estimates by averaging across uncertainty in the manifest variables. We extract the factor loadings for each country-year and store point estimates (medians) as well as the standard deviations from the BFA as the dimension variables presented in Table 1.⁹

In the data generation process, the sample included up to 117 countries worldwide with data since 2000, depending on data availability, and includes liberal and electoral democracies, as well as closed and electoral autocracies. Data for political trust in representative institutions, political trust in order institutions as well as satisfaction with democracy had been extrapolated for the years 2021 to 2023. The original data cover years up to 2020, but since these variables are imperfect versions of the respective concepts and come with measurement uncertainty, we decided to extrapolate the years 2021 to 2023 by using splines. In future iterations of the DRC Index, these extrapolated values can be replaced once the original data are updated in the original data source (Claassen, 2020; Valgarðsson et al., 2025).

Generally, higher factor loadings and lower uniqueness scores for individual indicators give stronger empirical evidence that the specific indicators relate more strongly to the underlying concept. All nine indicators of the macro-institutional attribute of resilience capacity load on a single dimension, though the uniqueness of the indicator “legislature questions officials in practice” indicates a weaker fit into the one-factor model. The second attribute of the resilience capacity—political actors—shows a moderate fit for the Anti-Pluralist Party Index, and a higher loading for the political polarization and political violence indicators. The uniqueness score for the Anti-Pluralist Party Index also indicates that a relatively large portion of the variance is unexplained by the political actors’ factor. In the civic culture and civil society attribute of resilience capacity, the factor analysis indicates that the six dimensions of this attribute load strongly on a single attribute. The uniqueness scores of the two indicators “power distributed by social group” and the “power distributed by gender dimensions” are somewhat higher. Nevertheless, the factor loadings indicate that they load on a single dimension. The five different survey items used to conceptualize the political community attribute of resilience capacity also load strongly on a single factor. Once again, the uniqueness scores indicate that a part of the variance is unexplained with this dimension, in particular for the “satisfaction with democracy” and the “trust in the police” latent variables. Although our approach for evaluating the resilience capacity has an exploratory nature, the findings provide overall moderate to strong empirical support regarding the content and construct validity of the resilience capacity measure.¹⁰

(4) Index Construction

We aggregate the factor scores of these attributes to three different indices.¹¹ First, we build an *additive Democratic Resilience Capacity Index* (aDRC) using each attribute equivalently. It is defined as:

$$\begin{aligned} aDRC = & \frac{1}{4} * \text{Macro-institutional attribute} + \frac{1}{4} * \text{Political actors' attribute} \\ & + \frac{1}{4} * \text{Civil society and civic culture attribute} + \frac{1}{4} * \text{Political community attribute} \end{aligned}$$

This additive index assumes that each attribute can compensate for lower values in other dimensions and thus conceptualizes the different attributes as mutually substitutable aspects of resilience capacity. In contrast, the *multiplicative Democratic Resilience Capacity Index* (mDRC) assumes that the different attributes are individually necessary conditions for resilience capacity. Thus, by using the multiplicative aggregation rule we combine information from all constitutive elements of resilience capacity. The mDRC Index is defined as:

$$\begin{aligned} mDRC = & \text{Macro – institutional attribute} * \text{Political actors' attribute} \\ & * \text{Civil society and civic culture attribute} * \text{Political community attribute} \end{aligned}$$

The question of whether to use multiplicative or additive indices has been debated in democracy measurement studies for decades, without conclusive results (see Bollen, 1980; Coppedge et al., 2020: 97–101; Coppedge and Reinicke, 1990; Goertz, 2006: 111–115; Munck, 2009; Przeworski et al., 2000; Treier and Jackman, 2008). Since both the necessary conditions and the substitutable logic have reasoned support, and since both have evidently the virtue of discriminating at different ends of the spectrum, we use the average between the two resilience capacity indices as our preferred solution to the aggregation of a complex concept, such as resilience capacity. The *Democratic Resilience Capacity* (DRC) *Index*¹² is thus constructed by averaging both indices as:

$$DRC = 0.5 * aDRC + 0.5 * mDRC$$

Figure 2 details the empirical distribution of the DRC Index, as well as the relationship between the different aggregation rationales. The left-skewed distribution of observations in the *multiplicative DRC Index* contrasts with a more balanced distribution of cases in the *additive DRC Index*, whereas the empirical distribution of the DRC Index lies somewhere between the two alternatives. Still, even here the distributions suggest that many democracies may exhibit substantial shortcomings in terms of their resilience capacity. In the Supplementary Appendix we also re-plotted Figure A1 with all regimes, including autocracies (cf. Section 4.2 for sampling rules).

Figure 3 shows six different country examples, namely, the United States, Russia, South Korea, France, Germany and Norway. Across these six countries Norway performs the best, followed by Germany, France and the United States. South Korea as a more recent democracy performs poorer in terms of its resilience capacity. Russia, which experienced autocratic hardening since the 2000s, performs the worst of all cases. To allow a test of face validity, Figure A2 to A5 in the Appendix presents the data for all countries in our sample.

Figure 4 shows the four dimensions of resilience capacity of the six countries presented in Figure 3. It indicates that the different dimensions represent separate categories of resilience. For example, the macro-institutional dimension, as well as the civil society and civic culture dimension in the United States, are relatively stable since 2000, with only marginal drops in recent years, while the political actors dimension and the political community dimension have decreased dramatically since the 2000s. This finding fits with the literature pertaining increasing political polarization and declining trust in democratic institutions in America (Druckman et al., 2023; Graham and Svobik, 2020). South Korea, in contrast, shows a growing political community dimension, which is now comparable to the US, while the macro-institutional and civil society and civic culture dimension varies more over time compared to the US.

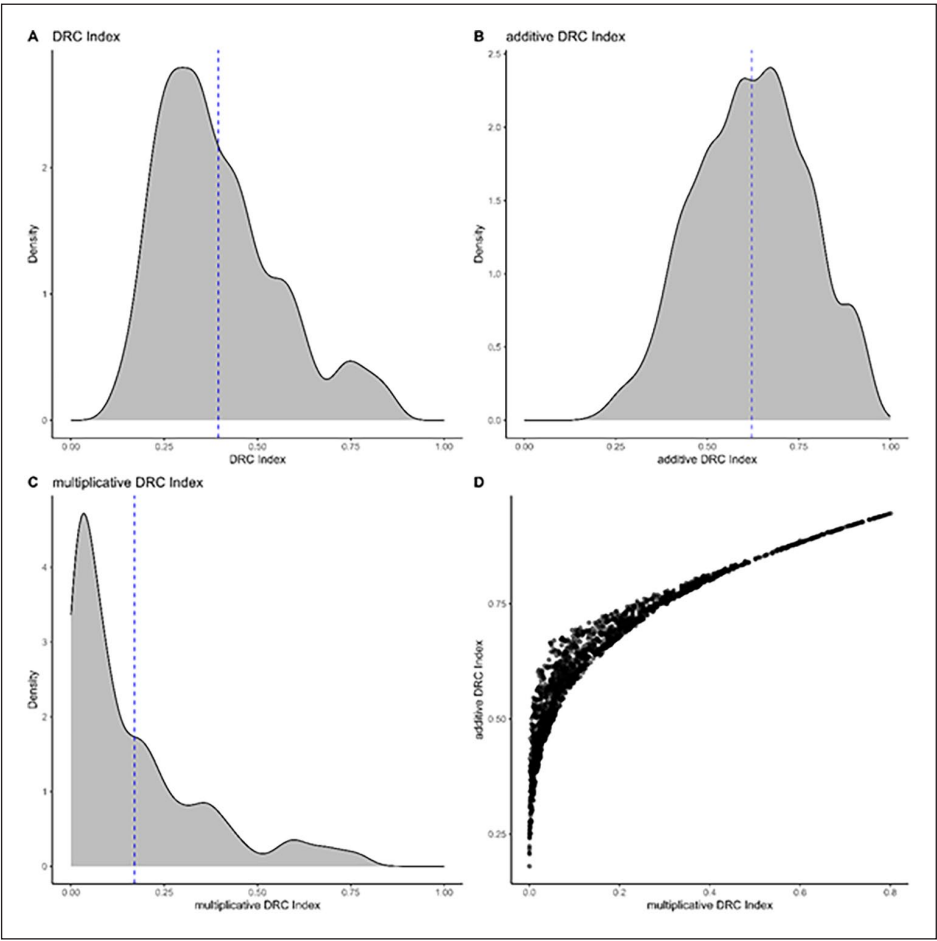


Figure 2. Aggregation to Democratic Resilience Capacity in Democracies. Blue vertical lines show the mean value for each DRC Index.

Exploring Democratic Resilience in the Twenty-First Century

In this section, we demonstrate the usefulness of our conceptualization of democratic resilience for empirical research through an exploratory study of resilience performance and resilience capacities in up to 103 democratic regimes across the world. We focus on the association between our DRC Index, its versions and the resilience performance of democracies since 2000.¹³ This time span covers a number of disruptive socioeconomic transformations and global crises which should have, perhaps, triggered a resilience response, such as the terrorist attacks on 9/11 and the Global War on Terrorism, the Global Financial Crisis (GFC) from 2007 to 2009, the COVID-19 pandemic (2020 to 2023) and the Ukraine War since February 2022. For this reason, and because of the common view that democracy in the early twenty-first century is increasingly challenged by autocratization (Lührmann and Lindberg, 2019; Møller and Skaaning, 2023), this period is particularly well suited to identify different levels and capacities of democratic resilience empirically.

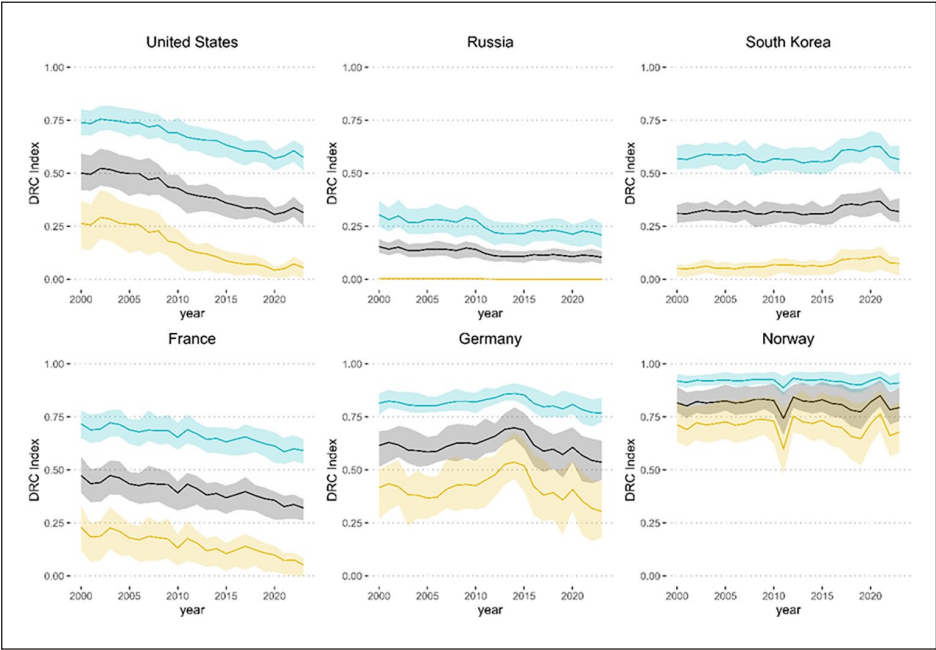


Figure 3. Country Examples and Democratic Resilience Capacity Indices. Black lines and areas show the DRC Index; yellow lines and areas show the multiplicative DRC Index; turquoise lines and areas show the additive DRC Index.

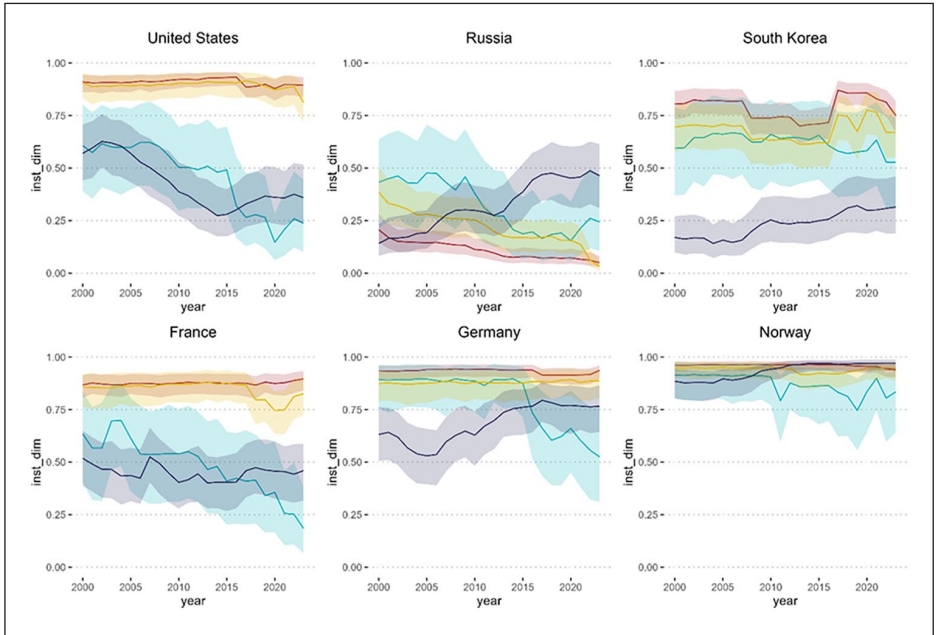


Figure 4. Country Examples and Democratic Resilience Capacity Dimensions. Red lines and areas show the macro-institutional dimensions; yellow lines and areas show the civil society and civic culture dimension; turquoise lines and areas show the political actors' dimension, and purple lines and areas show the political community dimension.

Table 3. Overview Outcomes.

Type	Number of episodes
Democratic breakdown	12
Democracies without breakdown	2
Regressed autocracies ^a	17
Censored outcomes	18
Democratic turnaround	
J-shape turnaround	7
U-shape turnaround	17
L-shaped turnaround	4
Total	77

^aCases classified as democracies by Boix, Rosato and Miller’s restrictive democracy criteria, but as autocracies in the ERT dataset.

As discussed before, we differentiate between three categories of democratic resilience: maintenance (“onset resilience”), resistance (the “breakdown resilience”) and bounce back (“turnarounds”). In a first step, we present a descriptive analysis of the distribution of these three categories and discuss some cases. In a second step, we analyze the association between the DRC Index, resilience against an autocratization onset and the breakdown resilience of democracies. In the third step, we analyze the association between resilience capacity and the ability of weakened democracies to bounce back, that is, if different types of democratic turnarounds are more likely when the resilience capacity is high. As an additional analytical approach, we differentiate between democracies that experienced a prolonged decline in democratic qualities but bounced back and those where autocratization was not followed by democratic turnarounds. To measure onset and breakdown resilience, as well as democratic turnarounds, we use the operationalization by Nord et al., (2025b) and the ERT dataset (Maerz et al., 2024a) in the V-Dem’s 14th version.

Descriptive Analysis

Table 3 shows the distribution of autocratization outcomes and democratic turnarounds for all ERT, which were at least one year active since 2000. Based on the dichotomous measure of democracy proposed by (Boix, Miller and Rosato, 2013) version 4, and extended until 2023 by the authors, we sample 103 democratic regimes. As mentioned above, democracies without an autocratization episode are considered to have onset resilience; those democracies that lack onset resilience but do not experience a breakdown, are considered as having breakdown resilience; and those that experienced a democratic turnaround are considered to have bounce-back resilience.

Overall, in this sample we have registered 77 autocratization episodes. From these 77 episodes, 28 episodes were democratic turnarounds, while the remaining 49 episodes did not result in a democratic turnaround (cf. Table 4). Of 49 episodes without onset resilience and without democratic turnaround, twelve resulted in a democratic breakdown. In two cases, democratic regression did not lead to democratic breakdown and it also was not followed by any positive movement toward more democratic qualities. In 17 episodes democratic qualities have been flawed enduringly (“regressed autocracies”), and another 18 episodes were still ongoing at the end of the coding period (“censored outcome”). Of

Table 4. Democratic Resilience Capacity and Outcomes.

Episode outcome	Mean	SD	Min	Max	No. cases
Democratic breakdown	0.307	0.113	0.169	0.480	12
No democratic breakdown	0.345	0.023	0.329	0.362	2
Outcome censored	0.336	0.108	0.138	0.578	18
Regressed autocracy	0.203	0.094	0.089	0.480	17
U-shaped turnaround	0.258	0.070	0.126	0.374	17
J-shaped turnaround	0.237	0.078	0.134	0.346	7
L-shaped turnaround	0.309	0.102	0.209	0.451	4

the 28 episodes in which democracies demonstrated some kind of bounce-back resilience, a total of seven episodes led to substantially higher democratic qualities (“J-shaped” turnaround).¹⁴ This diverse group includes Pakistan, Peru, Armenia, Lesotho, Madagascar and Nepal (twice). In another 17 cases, the initial autocratization process was followed by the restoration of the pre-episode democratic level (“U-turn”). Examples are South Korea between 2008 and 2017, Bangladesh from 2002 to 2010, and Brazil between 2016 and 2023. Finally, four cases yielded a substantially lower level of democratic qualities compared to the pre-episode levels (“L-shaped” turnarounds). This type of democratic turnaround was registered in Bolivia between 2006 and 2023, Thailand from 2013 to 2023, Tunisia 2013 to 2023 and Benin 2018 to 2023.

In Table 4, we calculated the average resilience capacity, the standard deviation and minimum and maximum values across the different episode outcomes. The results show that the resilience capacity was the highest in cases of censored outcomes (e.g. ongoing autocratization episodes) and in cases without democratic breakdown but without a democratic turnaround. The lowest average democratic resilience capacity was registered in J-shaped democratic turnarounds and in regressed autocracies.

Onset and Breakdown Resilience and Resilience Capacity

To investigate the association between resilience capacity and resilience performance more systematically, we adopt the research design by Boese et al. (2021) and test if resilience capacity is associated with more onset resilience and with more breakdown resilience. Boese et al. (2021) model two stages of democratic resilience empirically. For onset resilience, they use a standard onset model to examine the association between different determinants of democratic resilience. For the breakdown resilience of democracies, they use a standard bivariate probit model with non-random sample selection.

Overall, our sample of 103 democracies (as sampled by the BMR dataset; see above) includes 44 autocratization onsets since 2000. Similar to Boese et al. (2021), we use a probit model with Firth’s method of bias reduction as a standard onset model to estimate the onset resilience. Firth’s method of bias reduction was implemented to reduce small sample bias and to get a second-order unbiased estimator compared to the standard maximum likelihood estimator (see Beiser-McGrath, 2022; Firth, 1993; Kosmidis and Firth, 2021).¹⁵ Countries without onset resilience are coded as ones and democratic country-years in ongoing autocratization episodes are excluded in this onset model. Overall, we have 44 country-years (2.7% of all country-years) without onset resilience, which makes the reduction of small sample bias especially important.

To model breakdown resilience in the second stage, we use a standard bivariate probit model with non-random sample selection (Heckman, 1979).¹⁶ As in all selection models, the first stage models the probability that a country-year is in an autocratization episode, using the sample of 1922 country-years. Country-years in ongoing episodes are not excluded, logically. In the second outcome stage, we estimate the probability of a democratic breakdown using the subsample of country-years in an ongoing autocratization episode. The second stage outcome variable is coded as one for each episode-year in which the democracy broke down. The chosen estimators were used to implement models that are comparable to the models presented by Boese et al. (2021), while other estimators, such as rare events logits would be alternative estimators that can potentially be implemented.

In addition to the DRC Index, we also include other covariates. The selection of covariates builds on the study by Boese et al. (2021). We include measures of gross domestic product (GDP) per capita and GDP growth estimates from Fariss et al. (2022) to capture the level of economic development and the growth performance of democracies. We also include the population size and the average regional electoral democracy level using a six-fold geopolitical classification. We also include linear and quadratic time trends to account for global trends in resilience performance and global shocks. We employ regional dummies to control for unobserved time-invariant factors that are specific for some regions.¹⁷ All right-hand side variables, including the Democratic Resilience Capacity Index, are lagged by one year.

Table 5 presents the results of the standard onset model estimating the onset probability in Model 1 and two-stage Heckman results in Model 2. Tables A3 to A5 in the Supplementary Appendix present the results for the different versions of the DRC Index, as well as the different dimensions of resilience capacity.

To illustrate the substantive effects more intuitively, we simulate predicted probabilities. The plots in Figure 5 show how the probability of an autocratization onset varies with the resilience capacity of a country. With more resilience capacity a country's predicted probability of an autocratization onset decreases. With a resilience capacity of 0.19, which is the level of resilience capacity for Mexico in 2019, democracies have a predicted onset probability of 2.75% (95% confidence interval [CI]=[0.01, 0.064]). In contrast, democracies with a resilience capacity of 0.6, which is equivalent to Iceland in the late 2010s, have a predicted onset probability of 0.74% (95% CI=[0.0021, 0.0221]).

Figure 6 visualizes the second stage results of the Heckman selection model, as presented in Model 2. In the second stage, more resilience capacity is clearly related to a lower probability of a democratic breakdown. The predicted probability is much lower for countries with higher resilience capacity. For example, with a resilience capacity of 0.19, which is equivalent to the resilience capacity of Mexico in 2019, the predicted probability of a democratic breakdown is 1.08% in a country-year. For a resilience capacity of 0.4, the predicted probability of a democratic breakdown is 0.02%. In sum, comparing Figures 5 and 6 shows the importance of distinguishing empirically between different types of democratic resilience, that is onset, breakdown and, perhaps, bounce-back resilience.

Bounce-Back Resilience and Resilience Capacity

Finally, we analyze whether more resilience capacity prior to an autocratization onset increases the chances of a democratic bounce back after an initial democratic regression

Table 5. Main Results of Onset and Breakdown Resilience in 103 Democracies Since 2000.

	Model 1	Model 2	Breakdown
	Onset	First Stage	Onset
Intercept	-72.47 *** (15.79)	-37.833* (21.451)	32.024 (27.879)
Democratic Resilience Capacity	-1.28 * (0.59)	-4.333*** (1.169)	-6.462*** (2.065)
GDP pc log	-0.20 (0.13)	0.026 (0.045)	0.029 (0.057)
GDP growth	0.02 (0.04)	-0.062 (0.202)	-0.095 (0.21)
Population log	0.08 * (0.04)	0.067 (0.072)	-0.057 (0.086)
Regional democracy levels	0.29 (1.66)	0.386 (0.356)	0.469 (0.424)
Western Europe and North America	-0.39 (0.47)	0.409 (1.016)	0.83 (1.351)
Subsaharan Africa	0.18 (0.40)	0.316 (0.562)	0.655 (0.919)
Asia and Pacific	0.51 (0.33)	0.246 (0.702)	-3.478*** (0.718)
Eastern Europe and Central Asia	0.25 (0.23)	0.509 (0.516)	1.277 (0.833)
MENA	0.68 (0.74)	-1.574 (2.092)	-0.831 (3.255)
Year	1.24 *** (0.29)	0.644* (0.385)	-0.576 (0.503)
Year squared	-0.01 *** (0.00)	-0.003 (0.002)	0.003 (0.002)
AIC	373.86	1569.01	
BIC	443.98	1719.16	
Log Likelihood	-173.93	-757.5048	
Num. obs.	1626	1922	340 (27 breakdowns)

*** $p < 0.01$; ** $p < 0.05$.

or democratic breakdown. As a first descriptive test, we compare the resilience capacity 1 year before the democratic breakdown or the democratic turnaround, respectively, for those country-years that are in an ongoing autocratization episode. As we have shown before, more resilience capacity is statistically associated with more democratic resilience and a reduced probability of a democratic breakdown in an ongoing autocratization episode. Therefore, we also expect that countries with a democratic turnaround have a greater resilience capacity compared to the breakdown cases. However, this is not the case when comparing the median values in Figure 7.

In Figure 8, we plot the country trajectories of each of the 22 countries with one or more democratic turnarounds. The figure reveals that resilience capacity was relatively low for each country in which a democracy bounced back from a prior autocratization, which aligns with Figure 7.

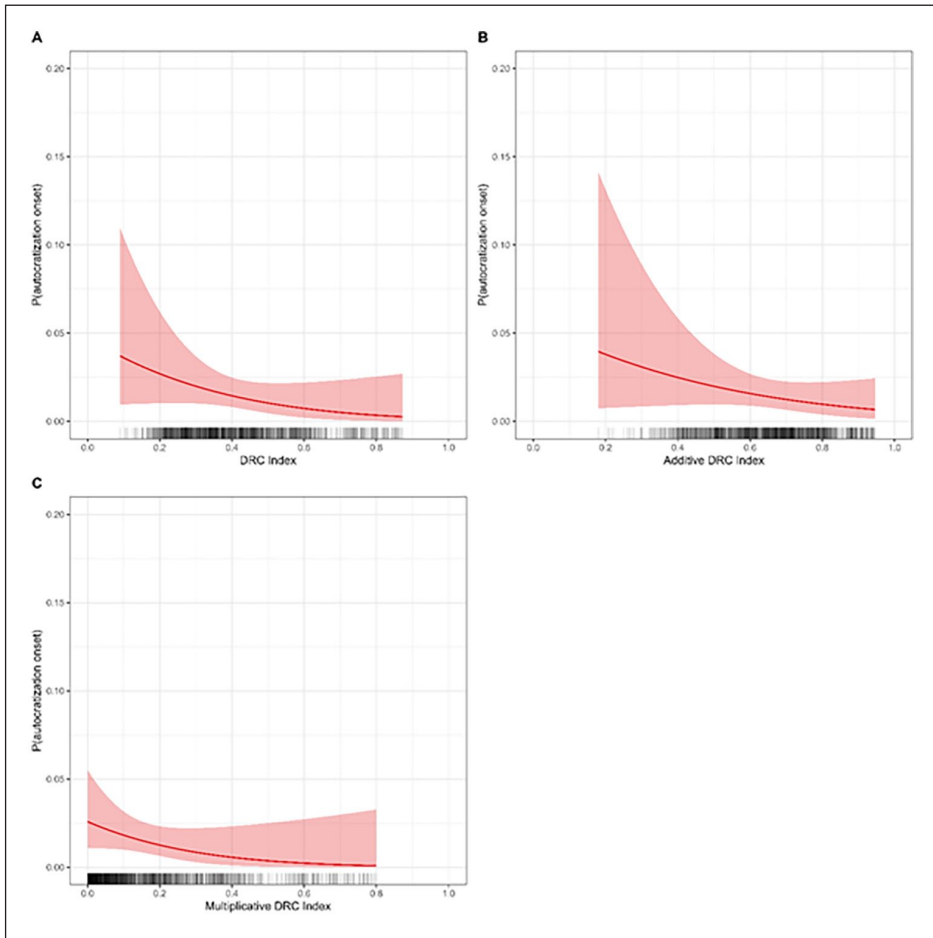


Figure 5. Predicted Probabilities of Onset Resilience Over Different Measures of Resilience Capacity. Estimates and 95% Confidence Intervals Are Based on Simulations.

While this suggests that resilience capacity as measured in this study and bounce-back resilience are not systematically related, a number of caveats are in order. First, the number of cases is too small for more systematic inferential statistical analysis. Second, the concept of democratic turnaround may be a flawed way to operationalize and measure democratic bounce-back resilience. Third, comparing the resilience capacity 1 year before a democratic breakdown or a democratic turnaround, respectively, could be insufficient because with a 1-year lag, the level of measured resilience capacity could already reflect the corrosive impact of ongoing democratic regression. In sum, we advise that future studies develop alternative operationalizations of the concept of bounce-back resilience.

Conclusion

This exploratory study aimed at contributing to empirical resilience research by introducing a novel conceptualization of democratic resilience and demonstrating how it can be applied in empirical research. We argued that in order to understand how democracies can

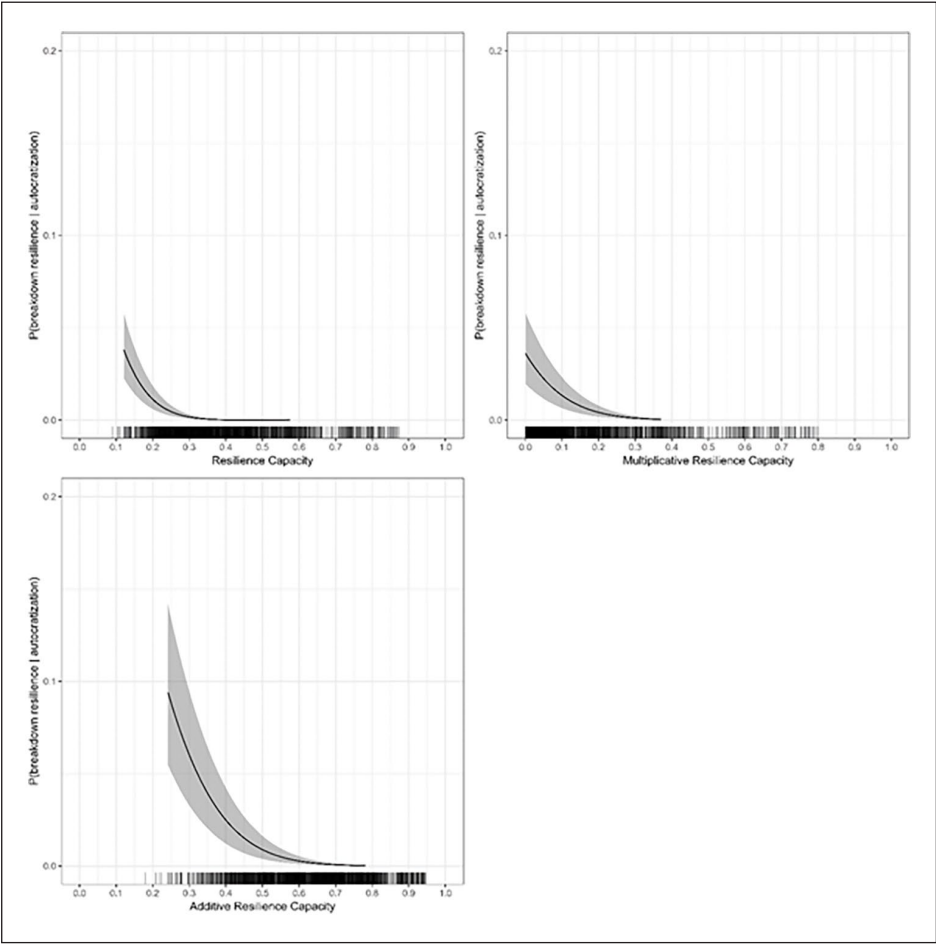


Figure 6. Predicted Probabilities of Breakdown Resilience Different Measures of Democratic Resilience Capacity. Estimates and 95% Confidence Intervals Are Based on Simulations From the Model Parameters.

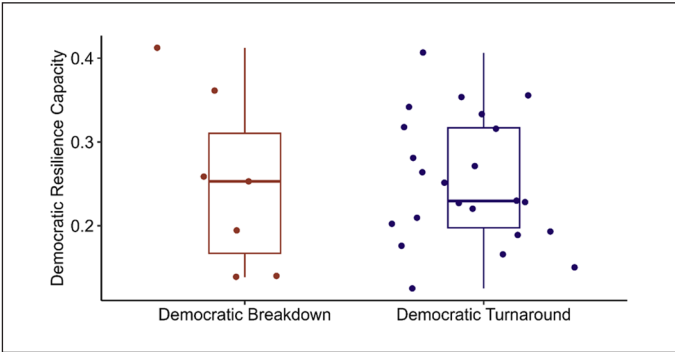


Figure 7. Democratic Resilience Capacity, One Year Before Breakdown or Turnaround.

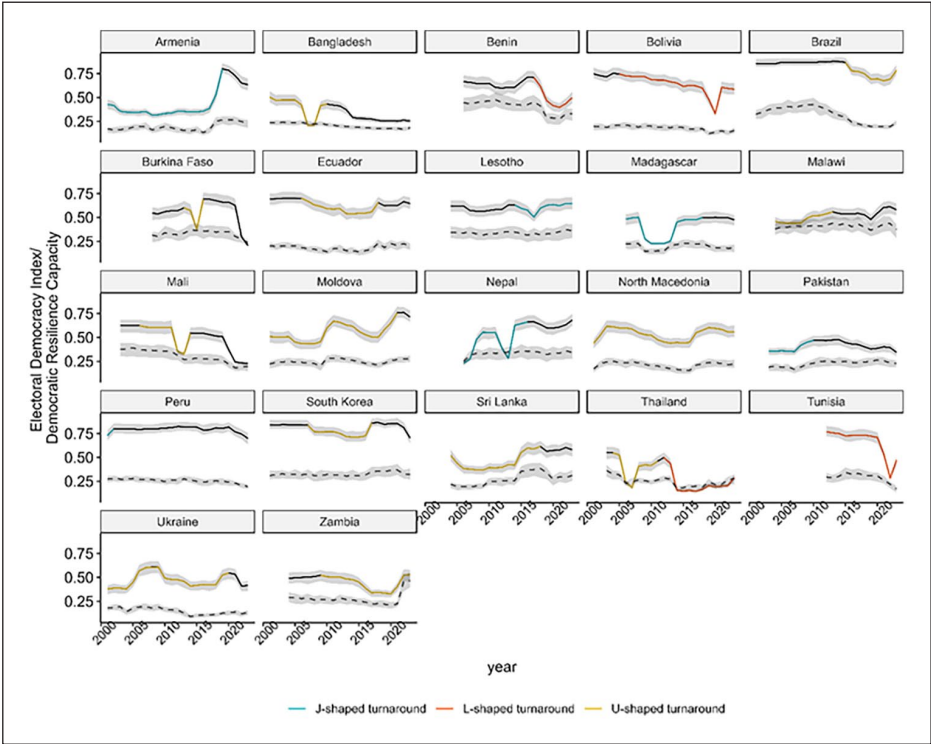


Figure 8. Electoral Democracy Index and Democratic Resilience Capacity for Every Democratic Turnaround Case.

counter autocratization, respond to democratic backsliding and bolster existing democratic institutions, it is necessary to differentiate between democratic resilience as performance (the level or degree of resilience a system exhibits at a given point in time) and resilience as capacity (a set of measurable characteristics within a given political system, which influence the level of democratic resilience). Based on the extant literature, we proposed measures for three stages of democratic resilience and also indicators for resilience capacity and developed an index of resilience capacity.

Our analysis of 103 democracies in the period 2000 to 2023 produced a number of findings which ought to motivate future research. We find that resilience capacity has a significant and substantial positive effect on both the probability that a democracy will be resilient against the onset of democratic backsliding and the breakdown of democracy. However, the probability that a political regime will bounce back from autocratization does not seem to be related to our indicators and dimensions of resilience capacity, though it remains to be seen if this is because of the operationalization of this type of democratic resilience, the indicators of resilience capacity or because the two phenomena are unrelated.

We propose four areas for expanding research into this nascent research agenda. First, expanding the temporal coverage of the empirical study will also require the production of new and better data. Second, the indicators identified as empirical referents of the concept of democratic resilience capacity are far from perfect. Therefore, a constructive dialogue and search for better indicators are warranted. Third, another area of follow-up research concerns the analysis of causal mechanisms, which is those causal processes

through which capacities impact the actual resilience performance of the system. So far, research on democratic regression and resilience has only rarely been combined with an approach that explicitly uses causal mechanisms as a theoretical or methodological concept (but see Croissant and Waldner, 2025; Lührmann et al., 2020; Massart et al., 2021). A comprehensive understanding of democratic resilience ought to understand “how” resilience contributes to the continuation, improvement, or recovery of democracy from stress, crisis or shock. Fourth, it is important to strengthen the connection between research about democratic resilience and the broader agenda of interdisciplinary resilience studies. In recent years, the world has witnessed a disturbing surge in the frequency and intensity of natural or human-made disaster and socioeconomic crises, exacerbated by climate change and geopolitical shifts. These events have left lasting impacts on communities, economies, cultural systems and ecosystems, demanding urgent attention and comprehensive solutions. In the face of such challenges, it is of crucial importance to understand how political, especially, democratic resilience can contribute to the facilitation of effective responses and foster societal resilience.

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Data Availability Statement

Research documentation and data that support the findings of this study are openly available at the Harvard Dataverse: <https://doi.org/10.7910/DVN/LY8FUM> or at Zenodo: <https://doi.org/10.5281/zenodo.15518083>. The scripts and data are also available on Code Ocean: <https://codeocean.com/capsule/2766645/tree/v1>. The reproduction materials contain all data that are necessary to computationally reproduce the results presented in this article and the supplementary appendix. The Democratic Resilience Capacity (DRC) Index is also available at Github: <https://github.com/LarsLott/DemocraticResilienceCapacityIndex>.

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Supplementary Information

Additional Supplementary Information may be found with the online version of this article.
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Notes

1. To our best knowledge, Bănică et al. (2021) are the first who have used the terminology of democratic resilience as “performance” and as “capacity.” Yet, the terms *ex ante/ex post* resilience that Volacu and Aligica (2023) introduced are similar.
2. The ERT is a dataset that utilizes V-Dem data on electoral democracy to capture episodes of regime transformation from 1900 to today.
3. For an overview, see the Routledge Handbook of Autocratization (Croissant and Tomini, 2024).
4. Data on the Anti-Pluralist Party Index capture anti-pluralism levels at the country-year level and is aggregated at the party-level based on the seat shares anti-pluralist parties gained in the legislature in the most recent national elections (see Angiolillo et al., 2025). We use the data from the Anti-Pluralism Index (Medzihorsky and Lindberg, 2024) to calculate the Anti-Pluralist Party Index. Data at the country-year level are extrapolated to the next elections within a 10-year period.
5. Valgarðsson et al. (2025) show that political trust can be differentiated between trust in representative institutions and trust in implementing institutions by applying a Bayesian dynamic latent trait model to more than 3000 cross-national surveys. By doing so, the authors find that “trust in representative institutions has generally been declining in recent decades, whereas trust in ‘implementing’ institutions has been stable or rising” (Valgarðsson et al., 2025: 1).
6. For the construction of the Democratic Resilience Capacity Index and its components, we use the full sample of all available country-years from 2000 onwards. We do not filter out non-democratic country-years. After the dataset construction, we analyze the resilience capacity of democracies, using the democracy definition from Boix et al. (2013).
7. We duplicate the baseline dataset 1800 times and assign the random draws from the posterior distribution of the latent variable to each country–year observation in the baseline dataset, by using the point estimates and standard deviations from the original datasets (V-Dem, V-Party, Claassen, 2020; Valgarðsson et al., 2025).
8. In order to capture posterior uncertainty, we run the Bayesian factor model 200 times (ITER) with 1800 different posterior draws from the variables and 10,000 sampling iterations (MCMC). We divide these runs into four groups, each with the same initial values, and for convergence purposes we treat each group as a separate chain to allow for a Gelman & Rubin diagnostic. This approach was also used by the V-Dem project to construct their manifest index variables.

9. In an additional test, we also test for a unidimensional factor model by using frequentist factor analysis and without accounting for the latent variable uncertainty. To do so, we use the point estimates and present the factor loadings and the uniqueness of the factors in Table A1 in the Supplementary Appendix. In sum, the frequentist factor analysis supports the BFA findings, but shows stronger empirical support for the dimensions' content validity: all indicators largely reflect the respective single underlying systematized concepts, that is, the four dimensions of resilience capacity.
10. A more systematized test of the construct and content validity of the Democratic Resilience Capacity Index can be implemented using the approach proposed by McMann et al. (2022). In this article, we discuss the content validity by assessing the BFA estimates. A critical assessment of the construct validity with other alternative measures of democracy can be found in the Supplementary Appendix Figures A6 to A11, where we provide insights regarding the data quality of the proposed resilience capacity index.
11. We construct the respective indices by loading the posterior distributions of the four components of resilience capacity and applying the different aggregation rules to each draw of the 900 posterior distributions. We then combine the posterior draws of the additive and multiplicative DRC Index, as well as our preferred solution of the DRC Index, by averaging across uncertainty in the manifest variables, namely using the median as point estimates and the standard deviation to estimate the 68% uncertainty intervals.
12. We provide a Codebook of the DRC Index in the Supplementary Appendix to ensure maximum transparency of the data generation process. In addition, all code for the data generation process is transparently documented at Harvard Dataverse and at CodeOcean Capsule.
13. Assuming the availability of data, which is challenging especially in respect to trust in representative and order institutions, the DRC Index could also be estimated for country-years before 2000.
14. Table A2 in the Supplementary Appendix presents the countries and the autocratization episodes and democratic turnarounds.
15. The standard onset models with bias reduction are estimated using the *brglm* R package (Kosmidis, 2021).
16. We use the *GJRM: Generalized Joint Regression Modeling* R package by Marra and Radice (2017).
17. We use these region dummies as a factor variable in the models.

References

- Andersson PF and Teorell J (2024) The Double-Edged Sword: How State Capacity Prolongs Autocratic Tenure but Hastens Democratization. *Journal of Conflict Resolution* 69 (6): 980–1004.
- Angiolillo F, Wiebrecht F and Lindberg SI (2025) Party Systems, Democratic Positions, and Regime Changes: Introducing the Party-System Democracy Index. *British Journal of Political Science* 55: e42.
- Bănică A, Eva M and Iașu C (2021) *The European Atlas of Resilience*. Editura Universităţii "Alexandru Ioan Cuza." Available at: <https://regroweu.uaic.ro/atlas.htm> (accessed 1 June 2025).
- Beiser-McGrath LF (2022) Separation and Rare Events. *Political Science Research and Methods* 10 (2): 428–437.
- Bernhard M, Hicken A, Reenock C, et al. (2020) Parties, Civil Society, and the Deterrence of Democratic Defection. *Studies in Comparative International Development* 55 (1): 1–26.
- Boese VA (2019) How (not) to Measure Democracy. *International Area Studies Review* 22 (2): 95–127.
- Boese VA, Edgell AB, Hellmeier S, et al. (2021) How Democracies Prevail: Democratic Resilience as a Two-Stage Process. *Democratization* 28 (5): 885–907.
- Bogaards M (2010) Measures of Democratization: From Degree to Type to War. *Political Research Quarterly* 63 (2): 475–488.
- Boix C, Miller M and Rosato S (2013) A Complete Data Set of Political Regimes, 1800–2007. *Comparative Political Studies* 46 (12): 1523–1554.
- Bollen KA (1980) Issues in the Comparative Measurement of Political Democracy. *American Sociological Review* 45 (3): 370–390.
- Chandler D and Coaffee J (eds) (2016) *The Routledge Handbook of International Resilience*. London: Routledge.
- Claassen C (2019) Estimating Smooth Country–Year Panels of Public Opinion. *Political Analysis* 27 (1): 1–20.
- Claassen C (2020) Does Public Support Help Democracy Survive? *American Journal of Political Science* 64 (1): 118–134.
- Collier D, Daniel Hidalgo F and Olivia Maciuceanu A (2006) Essentially Contested Concepts: Debates and Applications. *Journal of Political Ideologies* 11 (3): 211–246.
- Coppedge M (2017) *Eroding regimes: what, where, and when?* SSRN scholarly paper no. ID 3066677. Social Science Research Network. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4782726 (accessed 7 November 2017).
- Coppedge M and Reinicke WH (1990) Measuring Polyarchy. *Studies in Comparative International Development* 25 (1): 51–72.

- Coppedge M, Gerring J, Glynn A, et al. (2020) *Varieties of Democracy: Measuring Two Centuries of Political Change*. Cambridge: Cambridge University Press.
- Coppedge M, Edgell AB, Knutsen CH, et al. (2022) Causal Sequences in Long-Term Democratic Development and Decline. In: Edgell AB, Knutsen CH and Coppedge M, et al. (eds) *Why Democracies Develop and Decline*. Cambridge: Cambridge University Press, pp.215–261.
- Coppedge M, Gerring J, Knutsen CH, et al. (2024) *V-Dem Dataset V14* (Version 14) [Dataset]. Varieties of Democracy (V-Dem) Project. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4782726 (accessed 1 June 2025)
- Croissant A and Tomini L (eds) (2024) *The Routledge Handbook of Autocratization*. London: Routledge.
- Croissant A and Waldner N (2025) From capacity to performance. Pathways of democratic resilience in Asia. *Contemporary Politics*. Epub ahead of print 29 January 2025. DOI: 10.1080/13569775.2025.2459515.
- Druckman JN, Green DP and Iyengar S (2023) Does Affective Polarization Contribute to Democratic Backsliding in America? *The ANNALS of the American Academy of Political and Social Science* 708 (1): 137–163.
- Economist Intelligence Unit (EIU) (2025) *Democracy Index 2024*. London: The Economist. Available at: <https://www.eiu.com/n/campaigns/democracy-index-2024> (accessed 1 June 2025).
- Edgell AB, Wilson MC, Boese VA, et al. (2020) *Democratic Legacies: Using Democratic Stock to Assess Norms, Growth, and Regime Trajectories*. SSRN scholarly paper no. 3595957, 8 May. Social Science Research Network. Available at: <https://doi.org/10.2139/ssrn.3595957> (accessed 1 June 2025).
- Fariss CJ, Anders T, Markowitz JN, et al. (2022) New Estimates of Over 500 Years of Historic GDP and Population Data. *Journal of Conflict Resolution* 66 (3): 553–591.
- Firth D (1993) Bias Reduction of Maximum Likelihood Estimates. *Biometrika* 80 (1): 27–38.
- Folke C, Carpenter S, Walker B, et al. (2004) Regime Shifts, Resilience, and Biodiversity in Ecosystem Management. *Annual Review of Ecology, Evolution, and Systematics* 35: 557–581.
- Freedom House (2025) *Freedom in the World*. Washington, DC: Freedom House. Available at: https://freedom-house.org/sites/default/files/2025-02/FITW_World_2025_Feb.2025.pdf (accessed 1 June 2025).
- Fuchs D and Roller E (2018) Conceptualizing and Measuring the Quality of Democracy: The Citizens' Perspective. *Politics and Governance* 6 (1): 22–32.
- Geissel B (2023) *The Future of Self-governing, Thriving Democracies: Democratic Innovations By, with and for the People*. New York: Taylor & Francis.
- Gerring J, Bond P, Barndt WT, et al. (2005) Democracy and Economic Growth: A Historical Perspective. *World Politics* 57 (3): 323–364.
- Goertz G (2006) *Social Science Concepts: A User's Guide*. Princeton, NJ: Princeton University Press.
- Graham MH and Svolik MW (2020) Democracy in America? Partisanship, Polarization, and the Robustness of Support for Democracy in the United States. *American Political Science Review* 114 (2): 392–409.
- Haggard S and Kaufman RR (2016) *Dictators and Democrats. Masses, Elites, and Regime Change*. Princeton, NJ: Princeton University Press.
- Heckman JJ (1979) Sample Selection Bias as a Specification Error. *Econometrica* 47 (1): 153–161.
- Held D (2006) *Models of Democracy*. Cambridge: Polity Press.
- Holloway J and Manwaring R (2023) How Well Does “resilience” Apply to Democracy? A Systematic Review. *Contemporary Politics* 29 (1): 68–92.
- Houle C (2024) Inequality and Autocratization. In: Croissant A and Tomini L (eds) *The Routledge Handbook of Autocratization*. London: Routledge, pp.94–107.
- Inglehart R (1988) The renaissance of political culture. *American Political Science Review* 82(4): 1203–1230. <https://doi.org/10.2307/1961756> (accessed 1 June 2025).
- Kendall-Taylor A, Lindstaedt N, Frantz E, et al. (2019) *Democracies and Authoritarian Regimes*. Oxford: Oxford University Press.
- Knutsen CH and Teorell J (2024) Development and Decline of Democracies: The Role of Institutional Factors. In: Vatter A and Freiburghaus R (eds) *Handbook of Comparative Political Institutions*. Cheltenham: Edward Elgar Publishing, pp.379–394. Available at: <https://www.elgaronline.com/edcollchap/book/9781803929095/book-part-9781803929095-33.xml> (accessed 1 June 2025).
- Knutsen CH, Marquardt KL, Seim B, et al. (2024) Conceptual and Measurement Issues in Assessing Democratic Backsliding. *PS: Political Science & Politics* 57 (2): 162–177.
- Kosmidis I (2021) *brglm: Bias Reduction in Binomial-Response Generalized Linear Models* (Version 0.7.2) [Software]. Available at: <https://cran.r-project.org/web/packages/brglm/index.html> (accessed 1 June 2025)
- Kosmidis I and Firth D (2021) Jeffreys-Prior Penalty, Finiteness and Shrinkage in Binomial-Response Generalized Linear Models. *Biometrika* 108 (1): 71–82.

- Landau D and Dixon R (2020) Abusive Judicial Review: Courts against Democracy. *UC Davis Law Review* 53 (3): 1313–1388.
- Little AT and Meng A (2024) Measuring Democratic Backsliding. *PS: Political Science & Politics* 57 (2): 149–161.
- Lott L and Croissant A (2024) Measuring Autocratization. In: Croissant A and Tomini L (eds) *The Routledge Handbook of Autocratization*. London: Routledge, pp.49–64.
- Lührmann A and Lindberg SI (2019) A third wave of autocratization is here: what is new about it? *Democratization* 26(7): 1095–1113. <https://doi.org/10.1080/13510347.2019.1582029> (accessed 1 June 2025).
- Lührmann A, Marquardt KL and Mechkova V (2020) Constraining Governments: New Indices of Vertical, Horizontal, and Diagonal Accountability. *American Political Science Review* 114 (3): 811–820.
- McCoy J, Rahman T and Somer M (2018) Polarization and the Global Crisis of Democracy: Common Patterns, Dynamics, and Pernicious Consequences for Democratic Polities. *American Behavioral Scientist* 62 (1): 16–42.
- McMann K, Pemstein D, Seim B, et al. (2022) *Assessing data quality: online appendix*. Available at: <https://static.cambridge.org/content/id/urn:cambridge.org:id:article:S1047198721000279/resource/name/S1047198721000279sup001.pdf> (accessed 1 June 2025).
- Maerz SF, Edgell AB, Wilson MC, et al. (2024a) Episodes of Regime Transformation. *Journal of Peace Research* 61 (6): 967–984.
- Maerz SF, Schulte F and Trinn C (2024b) Autocratization and Political Conflict. In: Croissant A and Tomini L (eds) *The Routledge Handbook of Autocratization*. London: Routledge, pp.441–456.
- Marra G and Radice R (2017) *GJRM: Generalised Joint Regression Modelling* (Version 0.2-6.7) [Dataset]. Available at: <https://doi.org/10.32614/CRAN.package.GJRM> (accessed 1 June 2025).
- Massart T, Vos T, Egger C, et al. (2021) The Resilience of Democracy in the Midst of the COVID-19 Pandemic: Democratic Compensators in Belgium, the Netherlands and France Special Issue: Politics in the Low Countries in COVID-19 Times. *Politics of the Low Countries* 3 (2): 113–137.
- Medzihorsky J and Lindberg SI (2024) Walking the Talk: How to Identify Anti-pluralist Parties. *Party Politics* 30 (3): 420–434.
- Merkel W and Lührmann A (2021) Resilience of Democracies: Responses to Illiberal and Authoritarian Challenges. *Democratization* 28 (5): 869–884.
- Møller J and Skaaning S-E (2023) *Democratization and Autocratization in Comparative Perspective: Concepts, Currents, Causes, Consequences, and Challenges*. London: Routledge.
- Morlino L (2024) Past Legacies and Autocratization. In: Croissant A and Tomini L (eds) *The Routledge Handbook of Autocratization*. London: Routledge, pp.124–133.
- Munck GL (2009) *Measuring Democracy. A Bridge between Scholarship & Politics*. Baltimore, MD: The Johns Hopkins University Press.
- Munck GL and Verkuilen J (2002) Conceptualizing and Measuring Democracy: Evaluating Alternative Indices. *Comparative Political Studies* 35 (1): 5–34.
- Nord M, Altman D, Angiolillo F, et al. (2025a) *Democracy Report 2025: 25 Years of Autocratization—Democracy Trumped?* Gothenburg: University of Gothenburg: V-Dem Institute.
- Nord M, Angiolillo F, Lundstedt M, et al. (2025b) When Autocratization Is Reversed: Episodes of U-Turns since 1900. *Democratization*. Epub ahead of print 26 January 2025. DOI: 10.1080/13510347.2024.2448742.
- Przeworski A, Alvarez ME, Cheibub JA, et al. (2000) *Democracy and Development: Political Institutions and Well-Being in the World, 1950–1990*. Cambridge: Cambridge University Press.
- Riedl RB, Friesen P, McCoy J, et al. (2024) Democratic Backsliding, Resilience, and Resistance. *World Politics* 77: 151–177.
- Rogers P (2015) Researching Resilience: An Agenda for Change. *Resilience* 3 (1): 55–71.
- Rummel RJ (2002) *Power Kills: Democracy as a Method of Nonviolence*. London: Routledge.
- Sartori G (1976) *Parties and Party Systems: A Framework for Analysis*. Cambridge: Cambridge University Press.
- Skaaning S-E (2018) Different Types of Data and the Validity of Democracy Measures. *Politics and Governance* 6 (1): Article 1.
- Somer M, McCoy JL and Luke RE (2021) Pernicious Polarization, Autocratization and Opposition Strategies. *Democratization* 28 (5): 929–948.
- Sorsa V-P and Kivikoski K (2023) COVID-19 and Democracy: A Scoping Review. *BMC Public Health* 23 (1): 1668.
- Stollenwerk E, Börzel TA and Risse T (2021) Theorizing Resilience-Building in the EU's Neighbourhood: Introduction to the Special Issue. *Democratization* 28 (7): 1219–1238.

- Svolik M (2008) Authoritarian Reversals and Democratic Consolidation. *American Political Science Review* 102 (2): 153–168.
- Svolik M (2018) *When Polarization Trumps Civic Virtue: Partisan Conflict and the Subversion of Democracy by Incumbents*. SSRN scholarly paper no. 3243470, 3 September. Social Science Research Network. Available at: <https://doi.org/10.2139/ssrn.3243470> (accessed 1 June, 2025).
- Treier S and Jackman S (2008) Democracy as a Latent Variable. *American Journal of Political Science* 52 (1): 201–217.
- Treisman D (2020) Economic Development and Democracy: Predispositions and Triggers. *Annual Review of Political Science* 23 (1): 241–257.
- Valgarðsson V, Jennings W, Stoker G, et al. (2025) A Crisis of Political Trust? Global Trends in Institutional Trust from 1958 to 2019. *British Journal of Political Science* 55: e15.
- Vanhanen T (2000) A New Dataset for Measuring Democracy, 1810–1998. *Journal of Peace Research* 37: 251–265.
- Volacu A and Aligica PD (2023) Conceptualising Democratic Resilience: A Minimalist Account. *Contemporary Politics* 29 (5): 621–639.
- Walz K, Merten L, Rosenthal M, et al. (2025) From Latest Buzzword to Conceptual Framework: Unraveling the Complexities of Democratic Resilience. *Zeitschrift Für Vergleichende Politikwissenschaft*. Epub ahead of print 8 January 2025. DOI: 10.1007/s12286-024-00618-7.

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